Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2001-I-Q04] | Part A (Q1-Q5)

In ABC, $A = 60^\circ, B = 45^\circ$. The internal bisector AT meets BC and has length 24. Write the area as $a + b\sqrt{c}$ with positive integers and squarefree c. Find $a + b + c$.

三角形 ABC 中 $A = 60^\circ, B = 45^\circ$，内角平分线 AT 交 BC，且 $AT = 24$。面积写为 $a + b\sqrt{c}$，其中 c 无平方因子，求 $a + b + c$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2006-I-Q01] | Part A (Q1-Q5)

Quadrilateral $ABCD$ has $B = 90^\circ$, $AC \perp CD$, and $AB = 18, BC = 21, CD = 14$. Find its perimeter.

四边形 $ABCD$ 中 $B = 90^\circ, AC \perp CD, AB = 18, BC = 21, CD = 14$，求周长。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2007-II-Q03] | Part A (Q1-Q5)

Square $ABCD$ has side thirteen. Exterior points E, F satisfy $AE = CF = 12$ and $BE = DF = 5$, with E outside across AB and F outside across CD. Find EF^2.

边长十三的正方形 $ABCD$ 外，点 E 在 AB 外侧、点 F 在 CD 外侧，满足 $AE = CF = 12, BE = DF = 5$，求 EF^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2013-II-Q05] | Part A (Q1-Q5)

Points D, E trisect one side of an equilateral triangle ABC. Express $\sin\angle DAE = a\sqrt{b}/c$, with coprime positive a, c and squarefree b, and find $a+b+c$.

D, E 三等分等边三角形 ABC 的边 BC。将 $\sin\angle DAE$ 写为 $a\sqrt{b}/c$，a, c 为互质正整数、b 无平方因子，求三者和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2014-I-Q01] | Part A (Q1-Q5)

A shoelace joins eight eyelets on a 50-by-80 mm rectangle, four equally spaced on each long side, including the corners. It crosses the bottom width once, then its two strands each cross diagonally between successive eyelet levels three times. Each free end must extend another 200 mm. Find the shortest lace length.

八个鞋带孔位于五十乘八十毫米矩形两长边上，各边四孔等距且含顶点。鞋带先横跨底边一次，再由两股各斜跨至下一层孔，连续三层；两端另各留二百毫米。求最短鞋带长度。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2015-II-Q04] | Part A (Q1-Q5)

An isosceles trapezoid has parallel bases $\log 3, \log 192$ and altitude $\log 16$, using common logarithms. Its perimeter is $\log(2^p 3^q)$ for positive integers p, q. Find $p+q$.

等腰梯形的两底为常用对数 $\log 3, \log 192$，高为 $\log 16$。周长为 $\log(2^p 3^q)$，其中 p, q 为正整数。求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2018-I-Q04] | Part A (Q1-Q5)

In triangle ABC, $AB = AC = 10, BC = 12$. Interior side points $D \in AB, E \in AC$ satisfy $AD = DE = EC$. Reduce $AD = p/q$ and find $p + q$.

三角形 ABC 中 $AB = AC = 10, BC = 12$，边内部点 $D \in AB, E \in AC$ 满足 $AD = DE = EC$。将 AD 约为 p/q，求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2020-I-Q01] | Part A (Q1-Q5)

An isosceles triangle has $AB = AC$. Choose $D \in AC$ and $E \in AB$, both in the interiors of those sides, with $AE = ED = DB = BC$. Write $\angle ABC = m/n$ degrees in lowest terms. Determine $m + n$.

等腰三角形满足 $AB = AC$。在 AC, AB 的内部依次取点 D, E，使 $AE = ED = DB = BC$。若 $\angle ABC$ 的度数约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2021-II-Q05] | Part A (Q1-Q5)

For each $s > 0$, consider all obtuse triangles of area s having sides of lengths 4 and 10. The values of s for which at least one exists and all are congruent form $[a, b)$. Find $a^2 + b^2$.

对每个 $s > 0$，考虑面积为 s 且有两边长 $4, 10$ 的全部钝角三角形。使其非空且全部全等的 s 构成区间 $[a, b)$。求 $a^2 + b^2$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2023-II-Q03] | Part A (Q1-Q5)

In a right isosceles triangle ABC with right angle at A, an interior point P has $AP = 10$ and $\angle PAB = \angle PBC = \angle PCA$. Find the triangle area.

等腰直角三角形 ABC 的直角在 A。内点 P 满足 $AP = 10$ 且 $\angle PAB = \angle PBC = \angle PCA$。求三角形面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2003-I-Q10] | Part B (Q6-Q10)

Isosceles triangle ABC has $AC = BC$ and $C = 106^\circ$. Interior point M satisfies $\angle MAC = 7^\circ$ and $\angle MCA = 23^\circ$. Find $\angle CMB$ in degrees.

等腰三角形 ABC 满足 $AC = BC, C = 106^\circ$。内点 M 满足 $\angle MAC = 7^\circ, \angle MCA = 23^\circ$，求 $\angle CMB$ 度数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2003-II-Q07] | Part B (Q6-Q10)

In rhombus $ABCD$, the circumradii of ABD and ACD are 12.5 and 25. Determine the rhombus area.

菱形 $ABCD$ 中，三角形 ABD, ACD 的外接圆半径分别为 $12.5, 25$，求菱形面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2005-I-Q07] | Part B (Q6-Q10)

Quadrilateral $ABCD$ has $BC = 8, CD = 12, DA = 10$ and interior angles $A = B = 60^\circ$. If $AB = p + \sqrt{q}$ for positive integers p, q, find $p + q$.

四边形 $ABCD$ 中 $BC = 8, CD = 12, DA = 10$，内角 $A = B = 60^\circ$。若 $AB = p + \sqrt{q}$，其中 p, q 为正整数，求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2008-I-Q10] | Part B (Q6-Q10)

An isosceles trapezoid has longer base AD, base angles 60°, and diagonals $10\sqrt{21}$. A point E has $EA = 10\sqrt{7}, ED = 30\sqrt{7}$. If F is the perpendicular foot from C to AD and $EF = m\sqrt{n}$ with squarefree n, find $m + n$.

等腰梯形长底为 AD，底角六十度，对角线长 $10\sqrt{21}$。点 E 满足 $EA = 10\sqrt{7}, ED = 30\sqrt{7}$。$F$ 为 C 到 AD 的垂足，若 $EF = m\sqrt{n}$，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2013-I-Q09] | Part B (Q6-Q10)

Fold an equilateral paper triangle ABC of side twelve so A lands at T on BC with $BT = 9$. If the crease length is $m\sqrt{p}/n$, with coprime m, n and squarefree p, find $m + n + p$.

将边长十二的等边纸三角形折叠，使 A 落在 BC 上点 T，且 $BT = 9$。若折痕长为 $m\sqrt{p}/n$，m, n 互质、p 无平方因子，求三者和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2000-I-Q14] | Part C (Q11-Q15)

In isosceles ABC with $B = C$, choose $P \in AC, Q \in AB$ so $AP = PQ = QB = BC$. Let $r = \angle ACB / \angle APQ$. Find $\lfloor 1000r \rfloor$.

等腰三角形 ABC 满足 $B = C$，在 AC, AB 上取 P, Q，使 $AP = PQ = QB = BC$。令 $r = \angle ACB / \angle APQ$，求 $\lfloor 1000r \rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2003-I-Q12] | Part C (Q11-Q15)

Convex quadrilateral $ABCD$ has $A = C$, $AB = CD = 180$, unequal AD, BC, and perimeter 640. Find $\lfloor 1000 \cos A \rfloor$.

凸四边形 $ABCD$ 满足 $A = C, AB = CD = 180, AD \neq BC$，周长为 640，求 $\lfloor 1000 \cos A \rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2005-II-Q14] | Part C (Q11-Q15)

Triangle ABC has $AB = 13, BC = 15, CA = 14$. On BC, let $CD = 6$ and choose E with $\angle BAE = \angle CAD$. If $BE = p/q$ is reduced, find q.

三角形 ABC 中 $AB = 13, BC = 15, CA = 14$。$D, E$ 在 BC 上，$CD = 6$ 且 $\angle BAE = \angle CAD$。若 $BE = p/q$ 为最简分数，求 q。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2006-II-Q15] | Part C (Q11-Q15)

Real numbers satisfy $x = \sqrt{y^2 - 1/16} + \sqrt{z^2 - 1/16}$, $y = \sqrt{z^2 - 1/25} + \sqrt{x^2 - 1/25}$, $z = \sqrt{x^2 - 1/36} + \sqrt{y^2 - 1/36}$. If $x + y + z = m/\sqrt{n}$ with squarefree n, find $m + n$.

实数满足 $x = \sqrt{y^2 - 1/16} + \sqrt{z^2 - 1/16}$，$y = \sqrt{z^2 - 1/25} + \sqrt{x^2 - 1/25}$，$z = \sqrt{x^2 - 1/36} + \sqrt{y^2 - 1/36}$。若和为 $m/\sqrt{n}$ 且 n 无平方素因子，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2007-II-Q11] | Part C (Q11-Q15)

A radius-72 cylinder rolls without slipping along level ground, over a fixed parallel radius-24 cylinder, and back onto the ground. After exactly one full revolution it has its original circumferential contact point down again. The net horizontal displacement is $a\pi + b\sqrt{c}$, c squarefree. Find $a + b + c$.

半径七十二的圆柱在水平地面无滑动滚动，越过固定且平行的半径二十四圆柱后回到地面。恰转一整周，原周面接地点再次朝下，水平净位移为 $a\pi + b\sqrt{c}$，求 $a + b + c$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2008-II-Q14] | Part C (Q11-Q15)

For $a \geq b > 0$, the system $a^2 + y^2 = b^2 + x^2 = (a - x)^2 + (b - y)^2$ has a solution with $0 \leq x < a, 0 \leq y < b$. If the largest possible ratio is $\rho = a/b$ and $\rho^2 = m/n$ reduced, find $m + n$.

对 $a \geq b > 0$，方程组 $a^2 + y^2 = b^2 + x^2 = (a - x)^2 + (b - y)^2$ 有解满足 $0 \leq x < a, 0 \leq y < b$。若最大比值为 $\rho = a/b$，其平方为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2010-II-Q12] | Part C (Q11-Q15)

Two noncongruent integer-sided isosceles triangles have equal area and perimeter, and their base lengths are in ratio $8:7$. Find the least common perimeter possible.

两个不全等整边等腰三角形面积、周长均相等，底边长之比为 $8:7$。求最小的共同周长。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2010-II-Q14] | Part C (Q11-Q15)

In right triangle ABC, $\angle C = 90^\circ, \angle A < 45^\circ, AB = 4$. A point P on AB satisfies $CP = 1$ and $\angle APC = 2\angle ACP$. If $AP/BP = p + q\sqrt{r}$ with positive integers p, q and squarefree r, find $p + q + r$.

直角三角形 ABC 中，$\angle C = 90^\circ, \angle A < 45^\circ, AB = 4$。边 AB 上点 P 满足 $CP = 1$ 和 $\angle APC = 2\angle ACP$。若 $AP/BP = p + q\sqrt{r}$，p, q 为正整数、r 无平方因子，求 $p + q + r$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2018-II-Q12] | Part C (Q11-Q15)

Convex quadrilateral $ABCD$ has $AB = CD = 10, BC = 14, AD = 2\sqrt{65}$. Its diagonals meet at P and obey $[APB] + [CPD] = [BPC] + [APD]$. Find its area.

凸四边形 $ABCD$ 中 $AB = CD = 10, BC = 14, AD = 2\sqrt{65}$。对角线交于 P，且 $[APB] + [CPD] = [BPC] + [APD]$，求四边形面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2025-II-Q12] | Part C (Q11-Q15)

A simple nonconvex polygon $A_1 \cdots A_{11}$ has perimeter 20. For $i = 2, \ldots, 10$, triangle $A_i A_1 A_{i+1}$ has area 1 and angle at A_1 with cosine $12/13$. If $A_1A_2 + A_1A_{11} = (m\sqrt{n} - p)/q$ in reduced squarefree form, find $m + n + p + q$.

简单非凸十一边形 $A_1 \cdots A_{11}$ 周长为 20。对 $i = 2$ 至 10，三角形 $A_i A_1 A_{i+1}$ 面积为 1，且在 A_1 的角余弦为 $12/13$。若 $A_1A_2 + A_1A_{11} = (m\sqrt{n} - p)/q$ 为约简且根号内无平方因子的形式，求 $m + n + p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程